

Software Requirements Specification (SRS)

Functional Requirements

FR-1: The system shall allow the toll operator to provide vehicle input through live video streaming or recorded video files.

FR-2: The system shall automatically detect vehicles from the input video using an AI-based vehicle detection model.

FR-3: The system shall extract and identify vehicle details such as vehicle type and number plate.

FR-4: The system shall generate alerts when predefined detection rules or suspicious conditions are met.

FR-5: The system shall display generated alerts on the toll operator's dashboard.

FR-6: The system shall send alert notifications to the nearby traffic control room simultaneously.

FR-7: The system shall store alert information including vehicle details, owner name, owner contact number, toll ID, toll location, date, and time.

FR-8: The system shall support multiple toll plazas using a centralized database with a unique Toll ID.

FR-9: The system shall allow higher authorities (NHAI) to view alerts and vehicle details for monitoring and analysis purposes.

FR-10: The system shall maintain logs of detected vehicles and generated alerts for future reference and auditing.

Non-Functional Requirements

1. Performance Requirements

NFR-1: The system shall process video input and generate alerts in near real-time.

NFR-2: The system shall support concurrent operations from multiple toll plazas without performance degradation.

2.Security Requirements

NFR-3: The system shall restrict access to authorized users such as toll operators, traffic control room staff, and higher authorities.

NFR-4: The system shall ensure confidentiality of sensitive data, including vehicle owner information and contact details.

3.Reliability Requirements

NFR-5: The system shall operate continuously with minimal downtime.

NFR-6: The system shall ensure accurate vehicle detection and reliable alert generation.

4.Scalability Requirements

NFR-7: The system shall be scalable to support the addition of new toll plazas without major architectural changes.

5.Usability Requirements

NFR-8: The system shall provide a user-friendly web interface accessible through standard web browsers.

6.Maintainability Requirements

NFR-9: The system shall allow easy maintenance and upgrading of AI models and database components.

NFR-10: The system shall maintain proper error logs and system logs for monitoring and debugging purposes.